

genesis of Extremistan, but I insist that the “generator” of reality does not appear to obey them closely enough to make them helpful in precise forecasting. At least to judge by anything you find in the current literature on the subject of Extremistan. Once again we face grave calibration problems, so it would be a great idea to avoid the common mistakes made while calibrating a nonlinear process. Recall that nonlinear processes have greater degrees of freedom than linear ones (as we saw in [Chapter 11](#)), with the implication that you run a great risk of using the wrong model. Yet once in a while you run into a book or articles advocating the application of models from statistical physics to reality. Beautiful books like Philip Ball’s illustrate and inform, but they should not lead to precise quantitative models. Do not take them at face value.

But let us see what we *can* take home from these models.

Once Again, a Happy Solution

First, in assuming a scalable, I accept that an arbitrarily large number is possible. In other words, inequalities should not stop above some *known* maximum bound.

Say that the book *The Da Vinci Code* sold around 60 million copies. (The Bible sold about a billion copies but let’s ignore it and limit our analysis to lay books written by individual authors.) Although we have never known a lay book to sell 200 million copies, we can consider that the possibility is not zero. It’s small, but it’s not zero. For every three *Da Vinci Code*-style bestsellers, there might be one superbestseller, and though one has not happened so far, we cannot rule it out. And for every fifteen *Da Vinci Codes* there will be one superbestseller selling, say, 500 million copies.

Apply the same logic to wealth. Say the richest person on earth is worth \$50 billion. There is a nonnegligible probability that next year someone with \$100 billion or more will pop out of nowhere. For every three people with more than \$50 billion, there could be one with \$100 billion or more. There is a much smaller probability of there being someone with more than \$200 billion—one third of the previous probability, but nevertheless not zero. There is even a minute, but not zero probability of there being someone worth more than \$500 billion.

This tells me the following: I can make inferences about things that I do not see in my data, but these things should still belong to the realm of possibilities. There is an invisible bestseller out there, one that is absent from the past data but that you need to account for. Recall my point in [Chapter 13](#): it makes investment in a book or a drug better than statistics on past data